

专题三 展开定理

定义
Th

余子式与代数余子式的定义 在

$$|A| = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

(Handwritten red annotations: "i行" on the left, "j列" on top, and a red cross over element a_{ij})

中划去 a_{ij} 所在的第 i 行和第 j 列, 剩余的 $n-1$ 阶行列式称为 a_{ij} 的余子式, 记作 M_{ij} . 称 $(-1)^{\boxed{i+j}} M_{ij}$ 为

a_{ij} 的代数余子式, 记作 A_{ij} .

【评注】 a_{ij} 的余子式 M_{ij} 和代数余子式 A_{ij} 与第 i 行和第 j 列元素无关.

例: $|A| = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} 2 & 3 \\ 8 & 9 \end{vmatrix} = -(-6) = 6.$$

按行（或列）展开定理

(重要)

$$a_{i1}A_{j1} + a_{i2}A_{j2} + \cdots + a_{in}A_{jn} = \begin{cases} |A|, & \underline{\underline{i=j}} \\ 0, & \underline{\underline{i \neq j}} \end{cases}$$

$$a_{1i}A_{1j} + a_{2i}A_{2j} + \cdots + a_{ni}A_{nj} = \begin{cases} |A|, & i=j \\ 0, & i \neq j \end{cases}$$

【例 1.7】设 3 阶矩阵 A 的第 1 行元素为 1, 2, 3, $|A|$ 中第 2 行元素的余子式为 $a+1, a+2, a+3$, 则

$a =$ _____.

【详解】

$$\text{由 } 1 \times \overset{2+1}{(-1)}(a+1) + 2 \times \overset{2+2}{(-1)}(a+2) + 3 \times \overset{2+3}{(-1)}(a+3)$$

$$= -(a+1) - 2(a+2) - 3(a+3) = -2a - 6 = 0$$

$$\text{得 } a = -3.$$

【例 1.8】(2015, 数一) n 阶行列式

$D_n =$

$$\begin{vmatrix} 2 & 0 & 0 & \cdots & 0 & 2 \\ -1 & 2 & 0 & \cdots & 0 & 2 \\ 0 & -1 & 2 & \cdots & 0 & 2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 2 & 2 \\ 0 & 0 & 0 & \cdots & -1 & 2 \end{vmatrix} = \underline{\hspace{2cm}}.$$

【详解】

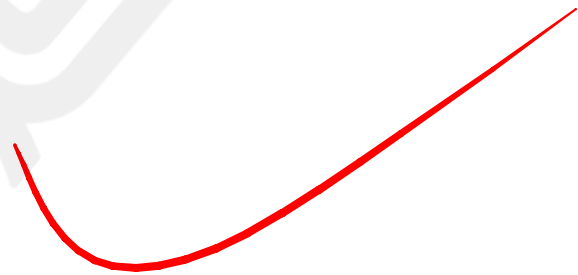
法一：展开第 n 行

$$D_n = 2 \times (-1)^{n+1} D_{n-1} + 2 \times (-1)^{n+1} (-1)^{n-1} = 2 D_{n-1} + 2$$

$$= 2(2D_{n-2} + 2) + 2 = 2^2 D_{n-2} + 2^2 + 2$$

$$= 2^{n-1} D_1 + 2^{n-1} + \dots + 2 = 2^n + 2^{n-1} + \dots + 2$$

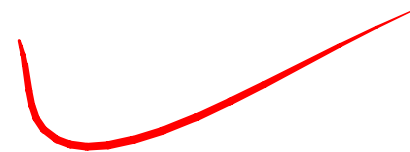
$$= \frac{2(1-2^n)}{1-2} = 2^{n+1} - 2$$



法二: 1, 3 列相加

$$\text{原式} = 2 \begin{vmatrix} 2 & 0 & 0 & \cdots & 1 \\ -1 & 2 & 0 & \cdots & 1 \\ 0 & -1 & 2 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & -1 \end{vmatrix} \xrightarrow{1+3} 2 \begin{vmatrix} 2 & 0 & 0 & \cdots & 1 \\ 0 & 2 & 0 & \cdots & 1 \\ 0 & 0 & 2 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 + \frac{2^{h+1}}{2^h} \end{vmatrix}$$

$$= 2 \times 2^h \left(1 + \frac{2^{h+1}}{2^h} \right) = 2^{h+1} - 2$$



【例 1.9】(2001) 设

$$|A| = \begin{vmatrix} 3 & 0 & 4 & 0 \\ 2 & 2 & 2 & 2 \\ 0 & -7 & 0 & 0 \\ 5 & 3 & -2 & 2 \end{vmatrix} = 5A_{41} + 3A_{42} - 2A_{43} + 2A_{44}$$

(I) 求第 4 行元素的余子式之和;

(II) 求第 4 行元素的代数余子式之和.

【详解】

注一: 定义 注二: 展开 7h (构造 13 列式)

$$(2) \quad \cancel{1}A_{41} + \cancel{1}A_{42} + \cancel{1}A_{43} + \cancel{1}A_{44}$$

$$= \begin{vmatrix} 3 & 0 & 4 & 0 \\ 2 & 2 & 2 & 2 \\ 0 & -7 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{vmatrix} = 0$$

$$(1) \quad M_{41} + M_{42} + M_{43} + M_{44}$$

$$= -A_{41} + A_{42} - A_{43} + A_{44}$$

$$= \begin{vmatrix} 3 & 0 & 4 & 0 \\ 2 & 2 & 2 & 2 \\ 0 & -7 & 0 & 0 \\ -1 & 1 & -1 & 1 \end{vmatrix} = -7(-1)^{3+2} \begin{vmatrix} 3 & 4 & 0 \\ 2 & 2 & 2 \\ -1 & -1 & 1 \end{vmatrix}$$

按第3行展开
第2列展开

$$= 7 \times (-4) = -28$$

○ 总结: (代数) 子式求和

$$(3) \quad 2A_{41} - 3A_{42} + \overset{+0 \cdot A_3}{A_{44}} =$$

$$\left| \begin{array}{ccc|c} \cdot & \cdot & \cdot & \\ \hline 2 & -3 & 0 & 1 \end{array} \right|$$

专题四 行列式的公式 (7个)

设 A, B 为 n 阶矩阵, 则

$$(1) |kA| = k^{\boxed{n}} |A|;$$

证明(数表)

$$\begin{aligned} \text{pr: } |kA| &= \left| k \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \right| \\ &= \begin{vmatrix} ka_{11} & \cdots & ka_{1n} \\ \vdots & & \vdots \\ ka_{n1} & \cdots & ka_{nn} \end{vmatrix} = k^n |A| \end{aligned}$$

$$A = \begin{pmatrix} 100 & 99 \\ 99 & 100 \end{pmatrix}$$

(2) $|AB| = |A||B|$; 积的行列式等于行列的积

$$(3) \quad |A^T| = |A|;$$

$\overset{=}{\text{列互换}}$

$$(4) \quad |A^{-1}| = \frac{1}{|A|};$$

$$\text{pr: } AA^T = A^T A = E$$

$$|AA^T| = |A| \underline{\underline{|A^T|}} = |E| = 1$$

$$\text{单位矩阵 } E = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix}$$

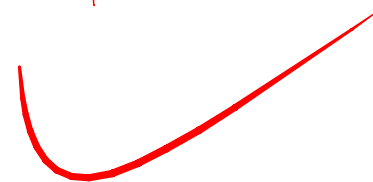
$$AE = EA = A$$

$$(5) |A^*| = |A|^{\boxed{n-1}};$$

pr. $A^* = |A| \cdot A^{-1}$

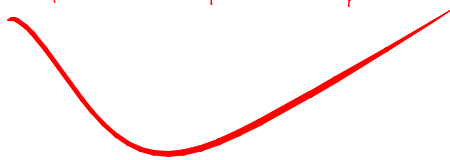
$$|A^*| = \left| \underbrace{|A|}_k A^{-1} \right| = |A| |A^{-1}| = |X|$$

$$= |A|^n |A^{-1}| = |A|^n \frac{1}{|A|} = |A|^{n-1}$$



(6) 设 A 的特征值为 $\lambda_1, \lambda_2, \dots, \lambda_n$, 则 $|A| = \prod_{i=1}^n \lambda_i$; pr: ok

(7) 若 A 与 B 相似, 则 $|A| = |B|$. pr: $B = P^{-1}AP$

$$|B| = |P^{-1}AP| = \cancel{|P^{-1}|} |A| \cancel{|P|} = |A|$$


行列式的求法

- (1) 数字行列式：利用重要行列式或展开定理；
(5回) (0n(最多))
- (2) 抽象行列式：利用行列式的性质或公式。
(1条) (1个)

二
联系

【例 1.10】 设 $\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2$ 均为 4 维列向量，且 $|\alpha_1, \alpha_2, \alpha_3, \beta_1| = m$ ， $|\alpha_1, \alpha_2, \beta_2, \alpha_3| = n$ ，则

$|\alpha_3, \alpha_2, \alpha_1, \beta_1 + \beta_2| =$

(A) $m + n$

(B) $-(m + n)$

(C) $n - m$

(D) $m - n$

【详解】

注：行列式性质

$$\begin{aligned} \text{原式} &= |\alpha_3, \alpha_2, \alpha_1, \beta_1| + |\alpha_3, \alpha_2, \alpha_1, \beta_2| = -|\alpha_1, \alpha_2, \alpha_3, \beta_1| + |\alpha_1, \alpha_2, \beta_2, \alpha_3| \\ &= n - m \end{aligned}$$

注二：行列法

$$\text{令 } d_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, d_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, d_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \beta_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ m \end{pmatrix}, \beta_2 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ -n \end{pmatrix}, \text{则}$$

$$T_{\text{总}} = -|d_1, d_2, d_3, \beta_1 + \beta_2| = - \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & m-n \end{vmatrix} = -(m-n) = n-m$$

【例 1.11】(2010, 数二、三) 设 A, B 为 3 阶矩阵, 且 $|A|=3$, $|B|=2$, $|A^{-1}+B|=2$, 则

$$|A+B^{-1}| = \underline{\hspace{2cm}}.$$

$$A+B=B+A$$

【详解】

$$\text{证: } |A+B^{-1}| = |AE+EB^{-1}|$$

$$= |ABB^{-1}+AA^{-1}B^{-1}| = |A(B+A^{-1})B^{-1}|$$

$$= |A| |A^{-1}+B| |B^{-1}| = 3 \times 2 \times \frac{1}{2} = 3$$

$$\text{解: } |A+B| = |EA+B|$$

$$= |B^{-1}B A + B^{-1}A^{-1}A| = |B^{-1}(B+A)A|$$

$$= |B^{-1}| |A+B| |A| = \frac{1}{2} \times 2 \times 3 = 3.$$

专题五 Cramer 法则

设线性方程组

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \cdots \quad \cdots \quad \cdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n \end{cases}$$

系数矩阵的行列式

$$D = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} \neq 0$$

则方程组有唯一解 $x_1 = \frac{D_1}{D}, x_2 = \frac{D_2}{D}, \cdots, x_n = \frac{D_n}{D}$, 其中

$$D_j = \begin{vmatrix} a_{11} & \cdots & a_{1j-1} & b_1 & a_{1j+1} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2j-1} & b_2 & a_{2j+1} & \cdots & a_{2n} \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & \cdots & a_{nj-1} & b_n & a_{nj+1} & \cdots & a_{nn} \end{vmatrix} \quad (j = 1, 2, \cdots, n)$$

推论 齐次线性方程组

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0 \\ \cdots \quad \cdots \quad \cdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = 0 \end{cases}$$

$$\Leftrightarrow r(A)=n$$

$$\Leftrightarrow r(A)<n$$

只有零解 $\Leftrightarrow D \neq 0$; 有非零解 $\Leftrightarrow D = 0$.

【例 1.12】设线性方程组

$$\begin{cases} (5-\lambda)x_1 + 2x_2 + 2x_3 = 0 \\ 2x_1 + (6-\lambda)x_2 = 0 \\ 2x_1 + (4-\lambda)x_3 = 0 \end{cases}$$

有非零解，则 $\lambda =$ _____.

【详解】

$$\text{由 } |A| = \begin{vmatrix} 5-\lambda & 2 & 2 \\ 2 & 6-\lambda & 0 \\ 2 & 0 & 4-\lambda \end{vmatrix} = (5-\lambda)(6-\lambda)(4-\lambda) - 4(6-\lambda) - 4(4-\lambda)$$

$$= (5-\lambda)(6-\lambda)(4-\lambda) - 8(5-\lambda)$$

$$= (5-\lambda) \{ (6-\lambda)(4-\lambda) - 8 \}$$

$$\lambda^2 - 10\lambda + 24 - 8$$

$$= (5-\lambda)(\lambda-2)(\lambda-8) = 0$$

∴ $\lambda = 2, 5, 8$.



【例 1.13】 设 a_1, a_2, a_3, a_4 互不相同, 则线性方程组

$$\begin{cases} x_1 + a_1 x_2 + a_1^2 x_3 + a_1^3 x_4 = b \\ x_1 + a_2 x_2 + a_2^2 x_3 + a_2^3 x_4 = b \\ x_1 + a_3 x_2 + a_3^2 x_3 + a_3^3 x_4 = b \\ x_1 + a_4 x_2 + a_4^2 x_3 + a_4^3 x_4 = b \end{cases}$$

的解为_____.

【详解】

$$D = \begin{vmatrix} 1 & a_1 & a_1^2 & a_1^3 \\ 1 & a_2 & a_2^2 & a_2^3 \\ 1 & a_3 & a_3^2 & a_3^3 \\ 1 & a_4 & a_4^2 & a_4^3 \end{vmatrix} = \prod_{1 \leq i < j \leq 4} (a_j - a_i)$$

$$D_1 = \begin{vmatrix} b & & & \\ b & & & \\ b & & & \\ b & & & \end{vmatrix} = b \prod_{1 \leq i < j \leq 4} (a_j - a_i)$$

$$D_2 = \begin{vmatrix} 1 & b & & \\ 1 & b & & \\ 1 & b & & \\ 1 & b & & \end{vmatrix} = 0, \quad D_3 = D_4 = 0$$

$$\text{解 } X_1 = \frac{D_1}{D} = b, \quad X_2 = \frac{D_2}{D} = 0, \quad X_3 = X_4 = 0.$$

$$\begin{pmatrix} b \\ 0 \\ 0 \\ 0 \end{pmatrix}$$